

English · [Dansk](#)

Formaldehyde cliff hardens; wood-panel substitution chooses pMDI and lignin, not silicate

The EU formaldehyde cliff on 6 August 2026 is firmly locked in by regulator and member state action, yet the substitution path the panel industry is visibly walking is reformulated MDI and lignin rather than inorganic silicate binders, narrowing the commercial window for silicate adhesives in wood-based products.

The consensus read on the EU formaldehyde cliff is that it opens a substitution window for low-toxicity binders, with silicate adhesives routinely cited as one of the chemistries that benefit. The cycle's actual momentum reads sharper and less favourable. Three regulatory clocks (REACH Entry 77, the German alignment repeal, and ECHA's measurement methodology) are accelerating in lock-step, but the supply-side response runs along two paths that bypass silicate: panel makers are transitioning to No Added Formaldehyde pMDI, and lignin-based binders are scaling beyond pilot. Bollerup Jensen's question for this cycle: does the silicate-adhesive window stay open long enough to capture share, or close as the panel chain locks in pMDI and lignin contracts before August?

Momentum across the topic

Signal	Direction	Evidence	Commentary
EU formaldehyde 0.062 mg/m3 cliff	ACCELERATING	Structural anchor; 6 August 2026 application date inside 70 days of cycle	Commission Regulation (EU) 2023/1464 will halve the EU formaldehyde limit to 0.062 mg/m3 from 6 August 2026, with no grace period (Notifix , April 2026).
ECHA measurement methodology published	ACCELERATING	Methodology released May 2025; 2 follow-on industry items in past 90 days	Compliance pipeline hardens as the canonical test method removes the last methodological uncertainty (EUWID Wood Products and Panels , June 2025).

Signal	Direction	Evidence	Commentary
German regulatory alignment locks in	ACCELERATING	ChemVerbotsV repeal scheduled for 7 August 2026; no softening signal in cycle	The EU 2023/1464 limits replace Germany's national ChemVerbotsV rule on 6 August 2026, halving the German benchmark rather than softening alignment (schadstoffrecht.de , January 2026).
Formaldehyde-resin chain consolidation	ACCELERATING	77.23% stake closed 13 March 2026, weeks before the cliff	Bondalti taking control of Ercros consolidates rather than exits Iberian formaldehyde-resin supply (EUWID Wood Products and Panels , March 2026).
Panel makers transitioning to NAF-MDI	ACCELERATING	Full MEDITE MDF range moved on 13 April 2026	MEDITE SMARTPLY chose No Added Formaldehyde MDI resin over silicate or bio-based alternatives for its compliance transition (MEDITE SMARTPLY , April 2026).
Lignin binders scale beyond pilot	ACCELERATING	New Zero product family flooring board announced May 2026	Koskisen and Stora Enso extended the NeoLigno Zero range to flooring, claiming industrial-scale production (Panels and Furniture Asia , May 2026).
Capital backing incumbent panel chemistry	ACCELERATING	EUR 146m EIB loan announced 18 March 2026 under REPowerEU framework	EIB backing Kronospan's energy upgrade entrenches the existing UF and pMDI panel model rather than substitute chemistry (European Investment Bank , March 2026).
Silicate geopolymers crosses structural threshold	ACCELERATING	French ATec for all superstructure use granted January 2026	Hoffmann Green's 0% clinker H-UKR cement (sodium-silicate-activated geopolymer) cleared for load-bearing applications (World Cement , January 2026).

Signal	Direction	Evidence	Commentary
CPR Working Plan and GWP declaration	ACCELERATING	First Working Plan published 15 December 2025; GWP mandatory from 8 January 2026	Embodied-carbon disclosure under the new Construction Products Regulation extends the pull on low-clinker, low-formaldehyde binders (European Commission DG GROW , December 2025).
EU sustainability reporting scope	FADING	Council Omnibus I adoption 24 February 2026; thresholds raised	CSRD coverage narrowing to over 1,000 employees softens the reporting-driven pull for substitute binders inside mid-tier panel producers (Council of the European Union , February 2026).

Pattern narrative

Substitution path closes around incumbents EVOLVING RISK

Four signals point the same way. MEDITE has moved its entire MDF range to fossil-route NAF-MDI rather than silicate or bio-based binders; the European Panel Federation puts the resin-cost step at 15 to 25% and names reformulated UF, MUF and NAF resins as the industry routes, not silicate ([Notifix](#), April 2026); Bondalti has consolidated the Iberian formaldehyde-resin chain six weeks before the cliff; and EIB capital is backing Kronospan's incumbent footprint. Read together, these are the closing-window evidence.

Three regulatory clocks chime at once **TRANSFORMATION DRIVER**

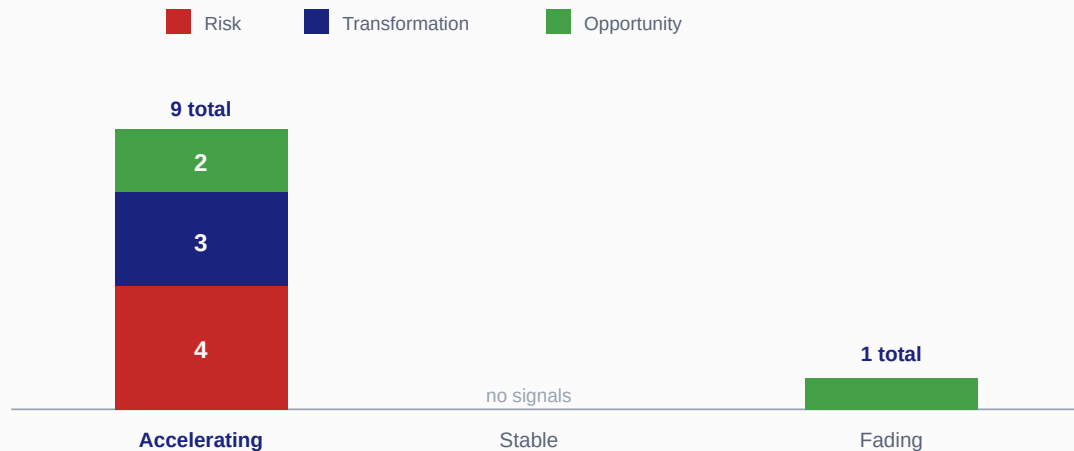
REACH Entry 77, the German ChemVerbotsV repeal and ECHA's measurement methodology move in lock-step, with the 0.062 mg/m³ limit applying on 6 August 2026 and the German rule sunseting the day after. Specialist 2026 commentary confirms the exemption list is too narrow to matter for indoor wood panels ([Analytical Group](#), March 2026). The new GWP declaration line under the Construction Products Regulation came into force in January, while the Council's Omnibus I raised CSRD thresholds in February: the cliff is real, but the sustainability-reporting tailwind has been pruned in the same cycle.

Silicate-activated geopolymer crosses a structural threshold

EMERGING OPPORTUNITY

The clearest positive signal is adjacent rather than panel-based. Hoffmann Green's H-UKR sodium-silicate-activated geopolymer received a French Technical Assessment for full superstructure use in January, and the company's H-P2A line positions silicate chemistry into industrial adhesives and fire protection ([Hoffmann Green Cement Technologies](#), evergreen). For a silicate producer, the read-across is that the ATec sets a regulatory template other silicate-activated systems can lean on. Bollerup Jensen is the only Danish liquid silicate producer ([Bollerup Jensen Group](#), evergreen), and no 2026 panel-adhesive launch is currently visible, which is itself a signal.

Momentum distribution across signals



Source basis: ten tracked signals from this cycle's sources.json, grouped by theme type and momentum direction.

Signals gaining momentum

1. **Wood-panel substitution to NAF-MDI.** A second visible incumbent transition in the next 60 days would force a commercial decision on whether to compete with pMDI suppliers or step back from panel adhesives.
2. **Formaldehyde-resin chain consolidation.** Further consolidation in Central European UF and MUF resins would compress the buyer set for silicate adhesives in the panel chain.
3. **Lignin Zero range expansion.** A second non-pilot lignin-binder flooring or panel deal in the next quarter would move lignin from credible to default substitute.
4. **Silicate-activated geopolymer in structural use.** A second Technical Assessment for an alkali-activated silicate binder in another EU jurisdiction would extend the Hoffmann Green template.
5. **ECHA methodology adoption.** First notified-body certifications under the new measurement protocol would lock in binder choices that are hard to reverse later.

Wild cards to watch

Lignin binder breakthrough captures the panel substitution share

DISRUPTIVE RISK *Potential impact: Very High*

Surprise characteristics: A bio-based binder previously confined to niches takes the post-cliff substitution share that silicate proponents had assumed.

Early warning indicators: Second large-volume NeoLigno or competing lignin product into a major panel maker; tier-1 retailer specifying lignin-bound panels.

Koskisen and Stora Enso's Zero extension is a credible breakthrough, with partners stating formaldehyde-free panels can work at industrial scale. If a second European panel major signs a lignin supply deal in the next two quarters, the substitution share for silicate adhesives narrows materially.

Silicate-activated geopolymer adhesive lands a flagship EU specification

DISRUPTIVE OPPORTUNITY *Potential impact: High*

Surprise characteristics: A national or major-city procurement spec writes sodium-silicate-activated geopolymer adhesive into a public-works or fire-protection standard.

Early warning indicators: Hoffmann Green or a peer extends H-UKR style approval into adhesive applications; a notified body issues a silicate-binder panel certificate under the new EU formaldehyde measurement methodology.

The Hoffmann Green H-UKR Technical Assessment for full superstructure use creates the regulatory template a silicate-activated adhesive line would need to enter major procurement spec language ([World Cement](#), January 2026). The trigger event would be a second silicate-binder ATec or equivalent national approval landing in another EU jurisdiction in the next two quarters.

Wild-card impact and surprise matrix



Source basis: two wild-card candidates from this cycle's evidence base; impact and surprise positions are analyst judgements traceable to the underlying sources.

Implications

The cycle reads the formaldehyde cliff as solid and the silicate-adhesive substitution window as narrower than consensus assumes. Worth monitoring: the next two NAF-MDI or lignin announcements from EU panel majors, which would allocate the substitution share before the cliff. Warrants preparation: a Bollerup Jensen position on whether to lead with a silicate-activated geopolymer adhesive line leaning on the Hoffmann Green H-UKR template, rather than competing head-on with pMDI. A decision point opens if a second silicate-binder Technical Assessment lands in another EU jurisdiction in the next quarter.

Bibliography

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

- **Tier 1** Commission presents new initiatives to support Europe's construction ecosystem. [European Commission DG GROW](#). Published 15/12/2025.
- **Tier 1** EIB backs Kronospan's green energy push with EUR 146 million loan. [European Investment Bank](#). Published 18/03/2026.

- **Tier 1** Council signs off Omnibus I simplification of sustainability reporting. [Council of the European Union](#). Published 24/02/2026.
- **Tier 2** ECHA publishes guidelines for measurement of formaldehyde emissions. [EUWID Wood Products and Panels](#). Published 13/06/2025.
- **Tier 3** Formaldehyd in Gebäuden: new EU limits from 2026 and the ChemVerbotsV transition. [Kanzlei Dr Elmar Liese \(schadstoffrecht.de\)](#). Published 20/01/2026.
- **Tier 3** Formaldehyde limits: what changes from 6 August 2026 under REACH Restriction 77. [Analytical Group](#). Published 25/03/2026.
- **Tier 3** Bondalti to buy a stake in Ercros at second attempt. [EUWID Wood Products and Panels](#). Published 25/03/2026.
- **Tier 3** EU REACH Revision (E05); MEDITE MDF range transitions to NAF-MDI. [MEDITE SMARTPLY](#). Published 13/04/2026.
- **Tier 3** EU to halve formaldehyde limit for wood panels starting August 2026. [Notifix](#). Published 23/04/2026.
- **Tier 3** Koskisen and Stora Enso expand formaldehyde-free wood panel range with new Zero product. [Panels and Furniture Asia](#). Published 25/05/2026.
- **Tier 3** Hoffmann Green validates H-UKR 0% clinker cement for superstructures. [World Cement](#). Published 12/01/2026.
- **Tier 4** H-P2A silicate-activated geopolymer cement for industrial adhesives. [Hoffmann Green Cement Technologies](#). Evergreen vendor page.
- **Tier 4** Silicates product page; Denmark's only liquid silicate producer. [Bollerup Jensen Group](#). Evergreen vendor page.

This Change Tracker is part of your Shaping Tomorrow Single-Topic Programme on **Silicate Adhesives**. To extend the programme to additional topics, add a named competitor, or move to a Sector-Exclusive Partnership, contact your analyst.

[Book your next-cycle discussion](#)

[Add a topic to your programme](#)